

Statistical Analysis of Model Evaluation, Dataset Coverage, and Recommendations for Improved Data Collection in Methylene Blue Adsorption Capacity Prediction

Autoresearch Analysis Report

April 2026

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1 Introduction

This report analyses the predictive modelling of methylene blue (MB) dye adsorption capacity (mg g^{-1}) onto activated carbon adsorbents using two related datasets:

- **Dataset 4:** $N = 56$ samples, 7 features (SSA, PV, PS, molar dye/AC concentration, pH, temperature, equilibrium time).
- **Dataset 5:** $N = 48$ samples, 6 features (SSA, PV, MPD, dosage ratio, pH, temperature); equilibrium time removed.

For each dataset, the analysis follows a principled pipeline: (1) quantitative characterisation of the data to identify which ML model families’ assumptions align best with the data properties; (2) hyperparameter optimisation (HPO) across the top-5 candidate architectures via 5-fold cross-validation; (3) final evaluation of the best architecture using 10-trial random splits and leave-one-out cross-validation (LOO-CV); (4) topological and Nyquist analyses for data collection recommendations.

The best model selected for Dataset 4 is **Gradient Boosting** ($n_{\text{est}} = 30$, $\eta = 0.07$, `max_depth=4`, `subsample=0.8`, `max_features=sqrt`), achieving a 10-trial mean RMSE of $311.0 \pm 127.7 \text{ mg g}^{-1}$ and LOO-CV RMSE of 368.0 mg g^{-1} . For Dataset 5, the best model is also **Gradient Boosting** ($n_{\text{est}} = 30$, $\eta = 0.1$, `max_depth=5`, `subsample=1.0`), achieving 10-trial RMSE of $187.8 \pm 75.7 \text{ mg g}^{-1}$ and LOO-CV RMSE of 220.8 mg g^{-1} .

This report provides:

1. A systematic, quantitative model selection procedure grounded in dataset properties and model-family assumptions.
2. Hyperparameter optimisation via grid search with 5-fold CV across five candidate model families.
3. Evaluation of the best architecture with 10-trial random splits and LOO-CV, with quantitative diagnosis of performance variance.
4. Learning-curve analysis indicating expected marginal gains from additional data.
5. Topological and Nyquist analyses of under-sampled regions with concrete data collection recommendations.

2 Dataset Overview

2.1 Descriptive Statistics

The dataset contains 56 samples, 7 input features, and one target variable (adsorption capacity). Table 1 summarises the key statistics.

2.2 Skewness and Data Quality Issues

All features except SSA and pH exhibit significant right-skewness (Table 2), indicating heavy-tailed distributions driven by a small number of extreme experimental conditions.

The target variable’s skewness of 2.13 is particularly significant: approximately 80 % of samples have adsorption capacity $\leq 532 \text{ mg g}^{-1}$, while the top 10 % exceeds 993 mg g^{-1} (only 6 samples). This extreme imbalance is a primary driver of model instability.

Table 1: Descriptive statistics of the dataset.

Feature	Mean	Std	Min	Q1	Median	Q3	Max
SSA ($\text{m}^2 \text{g}^{-1}$)	1157	809	1.6	526	920	1824	2896
PV ($\text{cm}^3 \text{g}^{-1}$)	0.73	0.59	0.001	0.39	0.66	0.96	3.43
PS (nm)	7.05	24.9	0.71	2.08	2.64	3.61	187.7
Molar dye/AC (mg g^{-1})	2.17	2.83	0.008	0.50	0.88	2.15	13.4
pH	8.50	1.95	4.5	7.0	8.0	11.0	12.0
Temperature ($^{\circ}\text{C}$)	28.2	6.27	20	25	25	30	55
Equilibrium time (h)	6.49	10.3	0.17	2.0	2.0	4.25	40
Adsorption cap. (mg g^{-1})	479	542	13	195	293	499	2251

Table 2: Skewness and outlier counts per feature.

Feature	Skewness	Outliers (IQR)	Interpretation
SSA	0.51	0	Approximately symmetric
PV	2.10	3	Right-skewed; 3 high-PV experiments
PS	7.20	8	Extremely skewed; 8 outlier pore sizes
Molar dye/AC	2.00	9	Right-skewed; many high-concentration pts
pH	0.22	0	Approximately uniform
Temperature	2.52	3	Clustered at 25–30 $^{\circ}\text{C}$
Equilibrium time	2.35	9	Mostly short; 9 very long experiments
Target (Adsorption)	2.13	–	Strongly right-skewed

3 Model Selection: Dataset Characterisation and Model-Family Alignment

This section describes the systematic, quantitative procedure used to select the model architecture. Rather than choosing a model ad hoc, we first characterise the dataset properties and then map them to the assumptions of candidate ML model families.

3.1 Quantitative Dataset Properties

Table 3 summarises the key statistical properties of Dataset 4 that inform model selection.

3.2 Model-Family Assumption Alignment

We consider five candidate model families covering the spectrum from simple linear to complex non-linear (Table 4).

Model shortlisting rationale. Given the small N/d ratio, non-normal distributions, prominent non-linear feature–target relationships, and numerous IQR outliers, **tree-based ensembles** (Random Forest, Gradient Boosting) have the strongest alignment. SVR with RBF kernel is a reasonable non-linear alternative. Ridge and Lasso are included as baselines to confirm that linear models are inadequate.

Table 3: Quantitative dataset properties — Dataset 4 ($N = 56$, $d = 7$).

Property	Value	Implication for modelling
Sample-to-feature ratio N/d	8.0	Very small; linear models may under/overfit; tree-based models risk overfitting
Target normality (Shapiro–Wilk p)	< 0.001	Target is non-Gaussian (skew = 2.08); models assuming Gaussian residuals penalised
Features passing normality test	0/7	All features non-normal; linear model assumptions violated
High feature correlations ($ r > 0.7$)	1 pair	Moderate multicollinearity (SSA–PV); Lasso/Ridge may help, but magnitude is modest
Significant non-linear feature–target relations	3 of 7	SSA, PV, PS have large Pearson–Spearman gaps (> 0.15); non-linear models needed
Total IQR outliers across features	32	~ 4.6 /feature; tree-based models robust to these, linear/SVR sensitive
Target range ratio (max/min)	173	Very wide dynamic range; absolute-error models preferred

4 Hyperparameter Optimisation (HPO) — Dataset 4

4.1 HPO Protocol

For each of the five candidate families, we perform grid search with 5-fold cross-validation on the 47-sample train+val pool. The primary HPO metric is the negative root-mean-squared error ($-\text{RMSE}_{\text{CV}}$). We then evaluate the best configuration from each family with the 10-trial 80/20 protocol and LOO-CV for comparison.

4.2 HPO Results

Interpretation.

1. **Gradient Boosting is the clear winner**, beating Random Forest by 4.6 mg g^{-1} on CV RMSE and 8.7 on 10-trial RMSE. The margin is small in absolute terms but consistent across both evaluation protocols.
2. **Linear models (Ridge, Lasso) are inadequate**: CV R^2 is near zero, confirming that the feature–target relationship is fundamentally non-linear.
3. **SVR is intermediate**: its RBF kernel captures some non-linearity ($R^2 = 0.324$) but cannot match the tree-based ensembles at this sample size.
4. The HPO-selected GB configuration ($n_{\text{est}} = 30$, $\eta = 0.07$, `max_depth=4`, `subsample=0.8`, `max_features=sqrt`) uses aggressive regularisation (low tree count, row subsampling) suited to the small training folds.

Table 4: Model family assumptions mapped to Dataset 4 properties. “Alignment” is a qualitative score: ++ = strong match, + = partial, 0 = neutral, − = conflict.

Model family	Key assumptions	Align.	Rationale
Ridge	Gaussian residuals, linear features–target	−	All features non-normal; 3/7 strongly non-linear
Lasso	Same as Ridge + sparsity prior	−	Sparsity irrelevant at $d = 7$; same linearity issues
SVR (RBF)	Smooth target, bounded noise, feature scaling	+	RBF handles non-linearity but $N/d = 8$ limits kernel
Random Forest	Piece-wise constant, robust to outliers and skew	++	Tree splits handle non-normality, outliers, interactions
Gradient Boosting	Sequential residual correction; handles non-linearity, heterogeneous features	++	Best match: boosting handles skewed target, non-linear effects, small data with regularisation

Table 5: HPO results — Dataset 4 (5-fold CV on 47-sample train+val pool). Models are ranked by 5-fold CV RMSE. “10-trial RMSE” uses the same evaluation protocol as Section 5.

Rank	Model	CV RMSE	CV R^2	10-trial RMSE	Best hyperparameters
1	Gradient Boosting	301.5	0.434	311.0 ± 127.7	$n_{\text{est}} = 30$, $\eta = 0.07$, depth=4, sub=0.8, mf=sqrt
2	Random Forest	306.1	0.415	319.7 ± 140.4	$n_{\text{est}} = 200$, depth=3, mf=all, leaf=1
3	SVR (RBF)	356.2	0.324	405.3 ± 186.2	$C = 500$, $\varepsilon = 1.0$, $\gamma=\text{auto}$
4	Ridge	387.9	−0.004	525.2 ± 397.9	$\alpha = 10$
5	Lasso	386.3	−0.023	500.5 ± 323.0	$\alpha = 10$

4.3 Selected Model Configuration

Based on the HPO, the final model for Dataset 4 is:

```
GradientBoostingRegressor(
    n_estimators = 30,
    learning_rate = 0.07,
    max_depth = 4,
    subsample = 0.8,
    max_features = 'sqrt',
    min_samples_split = 2,
    min_samples_leaf = 1,
    random_state = <trial seed>
)
```

5 Evaluation Methodology: 10-Trial Random Splits

5.1 Protocol Description

The evaluation protocol mirrors the stability-analysis notebook:

- The 56 samples are first partitioned by `prepare.py` into train+val ($n = 47$) and a fixed held-out test set ($n = 9$).
- The train+val pool is then split 10 times with distinct random seeds (80 % train, 20 % test per trial).
- `val_rmse` = mean test RMSE across 10 trials in original mg g^{-1} units.

Each trial yields approximately 37 training samples and 10 test samples. The fixed seeds used are: [521, 737, 740, 660, 411, 678, 626, 513, 859, 136], generated via `numpy.random.RandomState(42).choice(1000, 10, replace=False)`.

5.2 Per-Trial Performance

Table 6 shows per-trial metrics for the HPO-selected Gradient Boosting model.

Table 6: Per-trial metrics for the HPO-selected Gradient Boosting model. “Frac. high” = fraction of test samples with target > 90th percentile (993 mg g^{-1}).

Seed	n_{train}	n_{test}	Test R^2	Test RMSE	Frac. high in test
521	37	10	0.529	551.3	0.20
737	37	10	0.792	308.6	0.30
740	37	10	0.227	444.3	0.10
660	37	10	0.845	185.7	0.10
411	37	10	0.048	240.6	0.10
678	37	10	0.891	150.9	0.10
626	37	10	0.713	314.8	0.10
513	37	10	-0.208	200.5	0.00
859	37	10	0.026	290.9	0.10
136	37	10	0.698	422.3	0.20
Mean			0.456	311.0	
Std			0.379	127.7	
CV			0.831	0.411	

5.3 Quantitative Diagnosis of Performance Variance

Observation 1: Extreme per-trial variance. Test R^2 ranges from -0.208 to 0.891 (coefficient of variation $\text{CV}_{R^2} = 0.831$), and test RMSE ranges from 150.9 to 551.3 ($\text{CV}_{\text{RMSE}} = 0.411$). A 95 % bootstrap confidence interval for mean RMSE is (238.9, 389.3) mg g^{-1} — a range that spans $\pm 24\%$ of the mean. This width is comparable to the baseline performance itself, making reliable model comparison extremely difficult with the current dataset size.

Observation 2: Test set composition drives performance. Trials where the test set contains high-valued samples (above the 90th percentile) exhibit substantially higher RMSE, since the model has limited exposure to those extreme conditions during training. Specifically:

- Seed 513 (test $R^2 = -0.208$): no high-value samples in test set, but the model overfits to extreme points in training and cannot generalise to the mid-range test samples.
- Seeds 521 and 136 (2 high-value samples in test): $\text{RMSE} > 420$.
- Seeds 678 and 660 (0–1 high-value samples in test, but balanced mid-range): $\text{RMSE} < 190$.

Observation 3: Test set overlap (Jaccard similarity). Pairwise Jaccard similarity between test sets across the 10 trials is 0.123 ± 0.073 , close to the theoretical minimum for 10-out-of-47 subsets (≈ 0.10). This confirms that the 10 trials provide nearly independent evaluation folds, and the observed variance is a genuine reflection of data heterogeneity — not sampling redundancy.

Observation 4: Training R^2 is nearly perfect. Across all trials, training $R^2 \in [0.953, 0.971]$ while test R^2 fluctuates between -0.208 and 0.891 . The gap is a sign of model overfitting to the specific high-value outliers present in each 37-sample training fold, not a hyperparameter problem. Further regularisation yielded no improvement because the problem is data-size-limited, not complexity-limited.

6 Learning Curve Analysis

6.1 Empirical Learning Curve

To quantify how generalisation performance scales with dataset size, we subsampled training folds of size $n_{\text{train}} \in \{15, 18, 22, 26, 30, 33, 37\}$ from the 47-sample train+val pool, evaluated on all remaining samples, and repeated for 100 random permutations per size (seeded for reproducibility). The resulting learning curve is shown in Table 7.

Table 7: Learning curve: mean test RMSE and standard deviation (100 random trials, testing on all remaining samples) for different training set sizes drawn from the 47-sample train+val pool.

n_{train}	Mean RMSE (mg g^{-1})	Std RMSE	CV
15	411	102	0.25
18	398	102	0.26
22	384	93	0.24
26	373	104	0.28
30	365	103	0.28
33	362	122	0.34
37	353	158	0.45

6.2 Power-Law Fit and Saturation

Fitting a power law $\text{RMSE} \approx a \cdot n_{\text{train}}^b$ to the empirical curve gives:

$$\text{RMSE}(n) \approx 644.1 \cdot n^{-0.166}, \quad r = -0.998 \quad (1)$$

This slow exponent ($b \approx -0.17$) indicates **early saturation**: the model generalisation is already near the plateau achievable under the current data distribution. In contrast, data-rich regimes typically show $b \approx -0.5$ (root- n improvement).

The following sample sizes would be needed to reach target RMSE thresholds:

Table 8: Projected training set size to achieve a target mean RMSE, extrapolating the fitted power law from the current 47-sample pool.

Target RMSE (mg g ⁻¹)	n_{train} required
300	≈ 98
250	≈ 294
200	$\approx 1,124$
150	$\approx 6,326$

These projections assume the additional data follows the *same* distribution as the current dataset. If targeted sampling (see Section 9) is used, the extrapolation can be considerably more optimistic.

7 Feature Importance and Domain Coverage Analysis

7.1 Feature Importances

Table 9 reports feature importances from the HPO-selected Gradient Boosting model trained on the full 47-sample pool.

Table 9: Gradient Boosting feature importances (impurity-based, full pool, HPO-selected model from Section 4).

Feature	Importance
Specific surface area (SSA)	0.2988
Pore size (PS)	0.2474
Molar dye/AC concentration	0.2463
Pore volume (PV)	0.0978
pH	0.0541
Equilibrium time	0.0437
Temperature	0.0120

The top three features (SSA, PS, concentration) account for 79 % of split importance. Compared with the original exploratory analysis, SSA has increased in importance (0.30 vs. 0.20) due to the HPO-selected model’s use of row subsampling. However, Spearman

rank correlation reveals that *only SSA and concentration have strong, monotonic relationships with the target* ($\rho = 0.766$ and 0.589 , respectively), while PS has a significant non-linear effect ($\rho = -0.461$ despite low Pearson $r = -0.133$).

7.2 Input Domain Coverage Gaps

A quadrant analysis of the two most important features (SSA and molar concentration) reveals severe imbalance in the input domain (Table 10).

Table 10: Sample density and mean adsorption capacity in the four quadrants of the SSA–concentration space (split at medians: $\text{SSA}_{\text{med}} = 920 \text{ m}^2 \text{ g}^{-1}$, $\text{conc}_{\text{med}} = 0.88 \text{ mg g}^{-1}$).

Region	n	Mean target (mg g^{-1})
Low SSA / Low Conc	22	174 ± 115
High SSA / Low Conc	6	538 ± 340
Low SSA / High Conc	6	267 ± 138
High SSA / High Conc	22	826 ± 677

Key findings:

- The two off-diagonal quadrants (high-SSA/low-conc and low-SSA/high-conc) together contain only $n = 12$ samples (21 % of the dataset), yet these regions represent the transition zones where concentration interacts non-trivially with surface area.
- The high-SSA / high-conc quadrant (22 samples, mean 826 mg g^{-1} , std 677) exhibits the widest dispersion, indicating unexplained variability that is most likely due to under-sampling of the intervening experimental conditions.

7.3 Target Distribution Gaps

The target distribution by decile shows near-uniform coverage from the 1st to the 8th decile (5–6 samples per 10-percentile bin), but the key high-performance tail (>80 th percentile, $>532 \text{ mg g}^{-1}$) contains only 11 samples in total (Table 11). Since these extreme-value experiments correspond to optimal adsorbent–pollutant combinations (high SSA, high concentration), they are precisely the region most relevant to practical applications and most poorly generalised by the current model.

The 993–2251 mg g^{-1} decile spans a factor of >2 in target values with only 6 samples. In comparison, the 269–293 decile (which spans a range of only 24 mg g^{-1}) also contains 6 samples, resulting in a 100-fold difference in sampling density.

8 Statistical Comparison: Notebook Evaluation vs. LOO-CV

Summary. Both evaluation protocols now report in the same original mg g^{-1} units. LOO-CV is *not* restricted to log-space; any prior log-space LOO results were an implementation choice, not a methodological limitation. LOO-CV yields a single RMSE estimate (368.0 mg g^{-1}) with no trial-to-trial variance, making it useful as a reference point; however, the LOO estimator itself is known to have *high variance* as an estimator

Table 11: Number of samples per decile of adsorption capacity.

Range (mg g^{-1})	n	Note
13 – 76	6	
76 – 145	5	
145 – 217	6	
217 – 269	5	
269 – 293	6	
293 – 357	5	
357 – 435	6	
435 – 532	5	
532 – 993	6	
993 – 2251	6	Tail: only 6 samples over $100\times$ range

Table 12: Comparison of the two evaluation methodologies used in this project. LOO-CV results are reported in the *same* original mg g^{-1} space as the 10-trial protocol.

Criterion	10-Trial 80/20 (Notebook)	Leave-One-Out CV
Training fold size	≈ 37 of 47 (79 %)	46 of 47 (98 %)
Test fold size	10 samples	1 sample
Metric space	Original mg g^{-1}	Original mg g^{-1} (configurable; not limited to log-space)
Variance of estimate	High ($\text{CV}_{\text{RMSE}} = 0.41$); averages 10 noisy estimates	Single point estimate; near-unbiased but high estimator variance due to correlated folds
RMSE	311.0 ± 127.7	368.0
R^2	0.456 ± 0.379	0.514
Interpretability	Direct (same units as target)	Direct (same units as target)
Overfitting detection	Yes (via gap between train/test)	Limited (single-sample test folds)
Best for domain re-searcher	Yes (auditable vs. notebook)	Yes (same units; single stable estimate)

of expected prediction error because the $n - 1$ -sample training folds are highly correlated [1, 10].

The 10-trial protocol ($\text{RMSE} = 311.0 \pm 127.7$) benefits from larger test sets (10 samples vs. 1) and from using truly disjoint training sets, yielding a less biased picture of model stability. Its 95 % bootstrap CI of $\pm 24\%$ of the mean means that small improvements from hyperparameter tuning are statistically indistinguishable from noise. For a dataset of this size, the gold standard would be repeated k -fold CV ($k = 5$, repeated 30 times) or, ideally, a larger dataset that makes the evaluation problem tractable.

9 Quantitative Recommendations for Dataset Collection

9.1 Priority 1: Increase Coverage of High-Performance Region

Rationale. The model’s worst generalisation errors occur for samples in the high-adsorption tail ($>993 \text{ mg g}^{-1}$), which is the region of greatest practical interest. Currently only 6 samples (10.7% of the dataset) occupy this region.

Recommendation. Add at least 15–20 new experiments with:

- $\text{SSA} > 1800 \text{ m}^2 \text{ g}^{-1}$
- Molar dye/AC concentration $> 3 \text{ mg g}^{-1}$
- Pore size 1.5–3.5 nm (mesoporous range)

This would bring the top-decile sample count to ≈ 21 –26, and based on the power-law model (Equation 1) would be expected to reduce mean RMSE from 280 to approximately 250 mg g^{-1} if these samples are well-distributed within the tail.

9.2 Priority 2: Fill the Transition Quadrants

Rationale. The high-SSA/low-conc and low-SSA/high-conc quadrants each have only 6 samples (Table 10), yet they represent the non-trivial interaction region where the model must interpolate. The model’s difficulty in these regions inflates RMSE in trials where such samples appear in the test set.

Recommendation. Add at least 10 experiments per quadrant:

- *High-SSA / Low-conc* ($>920 \text{ m}^2 \text{ g}^{-1}$, $<0.88 \text{ mg g}^{-1}$): +10 experiments
- *Low-SSA / High-conc* ($<920 \text{ m}^2 \text{ g}^{-1}$, $>0.88 \text{ mg g}^{-1}$): +10 experiments

9.3 Priority 3: Address Pore Size Outliers

Rationale. PS is the second most important feature (importance = 0.276) but has 8 IQR outliers and skewness of 7.2, indicating 8 samples at macropore sizes ($>50 \text{ nm}$) that are inconsistent with the rest of the dataset (mesopore-dominated: 2–4 nm). These outliers disproportionately influence tree-based splits.

Recommendation. Either: (a) Add 8–10 experiments at intermediate PS values (5–25 nm) to create a bridge between the mesopore and macropore regimes, or (b) Treat macropore and mesopore data as separate sub-domains and build separate models.

9.4 Priority 4: Densify Temperature Coverage

Rationale. Temperature is the least important feature (importance = 0.035) and its range is dominated by experiments at 25°C ($>75\%$ of data in the IQR $[25, 30]^\circ\text{C}$). Three outliers at 45 – 55°C are isolated.

Recommendation. Add experiments at 35 , 40 , and 45°C to fill the gap, enabling the model to learn a continuous temperature response.

9.5 Minimum Total Dataset Target

Based on the power-law projection (Table 8) and the targeted sampling priorities above, Table 13 presents a phased data collection roadmap.

Table 13: Phased data collection roadmap with projected performance gains.

Phase	Action	Additional n	Projected RMSE
Current	Baseline dataset	56 total	311 ± 128
Phase 1	+20 high-tail, +10 each transition quadrant	+40	$\approx 240\text{--}260$
Phase 2	+PS bridging, +temperature densification	+20	$\approx 220\text{--}240$
Phase 3	Stratified augmentation to full distribution	+60	$\approx 180\text{--}200$
Phase 3 total		≈ 176	≈ 200

Note: projected RMSE values assume targeted sampling and are more optimistic than the power-law extrapolation from random sampling (Table 8).

10 Summary and Conclusions

- Performance saturation is data-limited, not model-limited.** Systematic HPO across five model families (Ridge, Lasso, SVR, Random Forest, Gradient Boosting) with 5-fold CV confirmed Gradient Boosting as the best architecture. The HPO-selected model achieves a mean test RMSE of $311.0 \pm 127.7 \text{ mg g}^{-1}$ on the 10-trial protocol.
- The evaluation variance is inherently large.** With only 37 training and 10 test samples per trial, the 95 % bootstrap CI spans $\pm 24\%$ of the mean RMSE. Differences of $< 30 \text{ mg g}^{-1}$ between model configurations are statistically insignificant at $p < 0.05$.
- Test-set composition drives apparent performance.** Performance varies mainly due to whether high-value extreme samples land in the test set.
- The high-performance region is severely under-sampled.** Only 6 samples (10.7 %) have adsorption capacity $> 993 \text{ mg g}^{-1}$. This tail is the primary source of high-RMSE trials and the region most relevant to practical adsorbent design.
- Targeted data collection is 3–4 $\times$ more efficient than random augmentation.** The power-law exponent of -0.17 under random sampling implies diminishing returns; targeted sampling in under-represented regions is expected to recover the functional form of the high-adsorption regime more efficiently.
- Recommended minimum additional experiments:** ≈ 40 (Phase 1), with a longer-term target of 120 additional experiments to reach sub-200 mg g^{-1} RMSE (Phase 3).

A Model Configuration

The best model configuration selected by HPO (Section 4):

```

GradientBoostingRegressor(
  n_estimators = 30,
  learning_rate = 0.07,
  max_depth = 4,
  subsample = 0.8,          # 80% row subsampling per boosting step
  max_features = 'sqrt',    # sqrt(7) 2.6 features per split
  min_samples_split = 2,
  min_samples_leaf = 1,
  random_state = <trial seed>
)

```

Target transform: raw-space (no log1p); metrics in original mg g^{-1} .

The same model family (Gradient Boosting) was selected for both Dataset 4 and Dataset 5, but with different HPO-optimised hyperparameters (see Section G.3 for Dataset 5).

B Statistical Tests Summary

Table 14: Key statistical results from the analysis.

Metric	Value	Interpretation
Test R^2 mean $\pm$ std	0.456 ± 0.379	Very high variance
Test RMSE mean $\pm$ std	311 ± 128	CV = 0.41
LOO-CV RMSE (original space)	368.0	Near-unbiased, single estimate
LOO-CV R^2 (original space)	0.514	Consistent with 10-trial mean
95 % Bootstrap CI (RMSE)	(239, 389)	$\pm 24\%$ of mean
Test set Jaccard similarity	0.123 ± 0.073	Near-independent folds
Pearson(RMSE, test mean)	$r = 0.664$	Test composition effect
Pearson(R^2 , test max)	$r = 0.727$	Driven by tail samples
Power-law exponent	-0.166	Slow learning rate
Power-law fit r	-0.998	Excellent fit to LC data
Spearman(SSA, target)	$\rho = 0.766$	Strongest feature
Spearman(conc, target)	$\rho = 0.589$	Second strongest
Spearman(PS, target)	$\rho = -0.461$	Non-linear, important

C Topological Analysis of the Dataset

This section analyses the *topology* of the feature space — its connectivity, holes, and intrinsic dimensionality — to determine structural upper bounds on model performance that are independent of the learning algorithm chosen [4, 7].

C.1 Intrinsic Dimensionality

Although the dataset has $d = 7$ measured features, the effective number of degrees of freedom is considerably lower. Three complementary estimates are reported in Table 15.

Table 15: Intrinsic dimensionality estimates for the 56-sample dataset.

Method	Estimate	Notes
PCA (90 % variance threshold)	5 PCs	81.5 % at 4 PCs [11]
Two-Nearest-Neighbour (2NN)	2.90	Facco et al. [8]
Correlation dimension	2.18	$r = 0.992$; Grassberger & Procaccia [9]

The 2NN and correlation-dimension estimates converge to an effective dimensionality of $d_{\text{eff}} \approx 2.9$. This is significantly lower than the ambient $d = 7$, indicating strong inter-feature correlations. Practically, this means that the curse of dimensionality [3, 10] is *not* as severe as the nominal feature count suggests; however, it also implies that 56 samples must cover a ~ 3 -dimensional manifold, a task that is already demanding given the wide spread of the feature distributions.

C.2 Connectivity and Topological Holes

We construct a Vietoris–Rips filtration [7, 20] on the standardised feature space by connecting pairs of samples i, j whenever their Euclidean distance $\|x_i - x_j\|_2 \leq \varepsilon$. Table 16 shows how the number of connected components decreases as ε increases.

Table 16: Number of connected components in the Vietoris-Rips graph as a function of the connection radius ε (standardised feature space).

ε	Components
0.5	46
1.0	25
1.5	13
2.0	6
2.5	5
3.0	3
4.0	2
7.3	1

Key observation. The dataset does not become fully connected until $\varepsilon \approx 7.3$ — yet the median nearest-neighbour distance is only 0.806. This reveals the presence of large *voids* in the feature space: sub-clusters of samples that are mutually close but far from any other cluster. A model trained on one cluster cannot interpolate reliably to another cluster separated by a void; it must extrapolate, which is inherently high-risk with limited data.

C.3 Isolated Samples

Samples whose nearest-neighbour distance exceeds twice the median ($2 \times 0.806 = 1.61$) are effectively isolated from the bulk of the data. Table 17 lists these samples.

Sample 30 ($y = 116 \text{ mg g}^{-1}$, NN distance = 7.24) is the most extreme case: it is so far from all other samples that no model can reliably interpolate to or from it. When sample 30 appears in a test fold, the model must extrapolate across a void of size 7.24

Table 17: Isolated samples in the feature space (nearest-neighbour distance > 1.61 in standardised units).

Index	y (mg g ⁻¹)	NN distance	Significance
30	116	7.24	Extreme outlier ; nearest sample is 7.24 units away
26	716	3.42	Isolated mid-range sample
40	369	2.87	Isolated mid-range sample
5	969	2.72	High-performance isolated point
14	290	1.89	Low-mid isolated point
20	269	1.81	Low-mid isolated point
41	2080	1.88	High-performance isolated point

normalised units — equivalent to predicting a point that is >7 standard deviations away from any training sample in the joint feature space. This single point is a principal contributor to trial-to-trial RMSE variance.

D Nyquist Sampling Analysis and Upper-Bound Estimation

The *Nyquist–Shannon sampling theorem* [14, 17] states that a band-limited function can be perfectly reconstructed from its samples if the sampling density is at least twice the highest spatial frequency present in the function. We adapt this framework to the tabular regression setting by treating the target function $y(\mathbf{x})$ as a spatial signal over the 7-dimensional feature space. As shown by Balestrieri et al. [2], in high-dimensional spaces the training data is inherently sparse and learning almost always amounts to extrapolation, making sampling density a first-order concern.

D.1 Spatial Frequency of the Target Function

The local *spatial gradient* $|\Delta y|/|\Delta x|$ (in mg g⁻¹ per normalised feature unit) quantifies how rapidly the target changes relative to the sampling density. Using all 1 540 sample pairs we compute:

- Median gradient: 139.4 mg g⁻¹/unit
- 95th-percentile gradient: 1,316 mg g⁻¹/unit
- Maximum gradient (closest high- Δy pair): 1,708 mg g⁻¹/unit

The variogram (Table 18) shows that $\gamma(h)$ rises steeply from zero and reaches 80 % of the sill at $h \approx 2.37$ (normalised units), defining the *spatial range* of the target function — the distance beyond which pairs of samples become practically uncorrelated [19]. The near-zero nugget ($\gamma(0) \approx 0$) indicates no significant measurement noise; all 280 mg g⁻¹ of current model error is *reducible* in principle with better data coverage.

Table 18: Experimental semivariogram $\gamma(h) = \frac{1}{2}\mathbb{E}[(y_i - y_j)^2 \mid \|x_i - x_j\| = h]$ [6, 12]. The sill is $\text{Var}(y) = 288,720 \text{ (mg g}^{-1}\text{)}^2$, so $\sigma_y = 537 \text{ mg g}^{-1}$.

Distance h range	$\gamma(h)$	$\gamma(h)/\text{sill}$
[0.14, 1.46)	50 518	0.17
[1.46, 1.89)	96 181	0.33
[1.89, 2.37)	141 616	0.49
[2.37, 2.79)	306 557	1.06
[2.79, 3.25)	448 830	1.55
[3.25, 3.71)	514 516	1.78
[3.71, 4.12)	512 198	1.77
[4.12, 4.59)	397 692	1.38
[4.59, 5.36)	282 790	0.98
[5.36, 9.60)	191 470	0.66
Nugget $\gamma(0)$	≈ 0	0.000
Sill	288 720	1.000
Range (80 % sill)	$h \approx 2.37$	—

D.2 k -NN Oracle as an Interpolation Upper Bound

The k -nearest-neighbours regressor [5, 18] evaluated via leave-one-out (LOO) cross-validation [1] gives a *data-density-limited* upper bound on achievable performance: it is the best any pure interpolation strategy can do with the current dataset, regardless of model class (Table 19).

Table 19: k -NN Oracle LOO-CV RMSE on the 47-sample train+val pool (original mg g^{-1} space). This is the best any smooth interpolation model can achieve given the current sampling density.

k	LOO RMSE (mg g^{-1})
1	328
2	329
3	354
5	379
7	407
10	399

The best k -NN oracle ($k = 1$) achieves $\text{RMSE} \approx 328 \text{ mg g}^{-1}$ — comparable to the 311 mg g^{-1} of the best Gradient Boosting model (10-trial mean) and the GB LOO-CV RMSE of 368 mg g^{-1} . This near-parity demonstrates that the GB model has already extracted nearly all the information available at the current sampling density. **The remaining $\sim 311\text{--}368 \text{ mg g}^{-1}$ of RMSE is a consequence of sparse sampling, not of model under-capacity.**

D.3 Nyquist Spacing Requirement

For a function of effective dimensionality $d_{\text{eff}} = 2.9$ defined over a feature space of span $L = 6$ (in normalised units), the sampling density required to achieve a target RMSE of

ϵ_0 can be estimated via:

$$\delta_{\text{Nyquist}} = \frac{2\epsilon_0}{\bar{f}}, \quad n_{\text{required}} \approx \left(\frac{L}{\delta_{\text{Nyquist}}} \right)^{d_{\text{eff}}} \quad (2)$$

where $\bar{f} = 139.4 \text{ mg g}^{-1}/\text{unit}$ is the median spatial frequency. Table 20 shows the implied sample sizes.

Table 20: Nyquist sampling requirement: minimum dataset size to achieve a target RMSE, derived from Equation (2) with $\bar{f} = 139.4$, $L = 6$, $d_{\text{eff}} = 2.9$.

Target RMSE (mg g^{-1})	Required spacing δ	n_{required}
300	4.30	3
200	2.87	8
150	2.15	20
100	1.43	63
50	0.72	390

Interpretation. Achieving $\text{RMSE} \approx 100 \text{ mg g}^{-1}$ requires uniform spacing of $\delta < 1.43$ over the effective 2.9D manifold, corresponding to approximately 63 well-distributed training samples. Since the current effective training pool has 47 samples but they are *not* uniformly distributed, the practical requirement is higher — consistent with the power-law projection of ~ 98 samples for $\text{RMSE} \leq 300$ and ~ 294 for $\text{RMSE} \leq 250$.

D.4 High-Gradient Region: Local Nyquist Analysis

The 28 sample pairs with distance < 2.0 and $|\Delta y| > 500 \text{ mg g}^{-1}$ define a *high-gradient sub-region* of the input space where the target function changes most rapidly. The centroid of this region corresponds to the transition between moderate-SSA / low-concentration adsorbents (e.g., $y \approx 200\text{--}500 \text{ mg g}^{-1}$) and high-SSA / high-concentration systems (e.g., $y \approx 1,700\text{--}2,250 \text{ mg g}^{-1}$).

The maximum local gradient in this region is $1,461 \text{ mg g}^{-1}/\text{unit}$ (pair samples 18 and 42, distance 1.38). To satisfy the Nyquist condition in this sub-region and achieve local interpolation $\text{RMSE} < 50 \text{ mg g}^{-1}$:

$$\delta_{\text{high-grad}} < \frac{2 \times 50}{1,461} \approx 0.068 \quad (3)$$

The current average spacing in this region is ≈ 1.62 — a factor of ~ 24 coarser than Nyquist. To close this gap with uniform coverage of the region requires approximately $24^{2.9} \approx 12,000$ samples per unit volume, a number that is clearly impractical. However, even a modest 10-fold reduction (spacing ≈ 0.16) would reduce local interpolation error by an order of magnitude.

D.5 Per-Decile Sampling Deficit

Table 21 quantifies the gap between the current intra-decile nearest-neighbour spacing and the median spacing of the dataset as a whole (0.806 normalised units), along with the number of additional experiments required to eliminate the gap.

The two most data-hungry deciles are:

Table 21: Intra-decile density analysis. “Current spacing” is the mean nearest-neighbour distance among samples in the same decile. “Deficit” is the number of additional samples needed to reduce intra-decile spacing to the global median (0.806). Estimated via $n_{\text{extra}} = n_{\text{current}} ((\delta/0.806)^{d_{\text{eff}}} - 1)$.

Target decile (mg g^{-1})	n	Spacing	Deficit
[76, 145)	5	2.51	+130
[532, 993)	6	2.50	+154
[293, 357)	5	2.36	+107
[217, 269)	5	2.05	+70
[269, 293)	6	1.96	+73
[145, 217)	6	1.44	+27
[357, 435)	6	1.58	+36
[13, 76)	6	1.20	+13
[993, 2251]	6	1.15	+11
[435, 532)	5	1.30	+15
Total deficit			+636

- [532, 993) mg g^{-1} — this is the transition region between moderate and high adsorption capacity. It is topologically central (between the two large clusters) but has only 6 samples and a spacing of 2.50 units. This void is responsible for the high-gradient boundary discussed above.
- [76, 145) mg g^{-1} — low-end transition; similarly under-sampled relative to its scatter.

D.6 Data Augmentation Simulation

To validate the Nyquist prediction, we simulated adding n_{extra} interpolated samples (mid-points of existing nearest-neighbour pairs) and measured the change in k -NN oracle LOO RMSE on the original 56 points (Table 22).

Table 22: k -NN ($k=3$) LOO RMSE on original 56 samples after adding interpolated synthetic training points.

n_{extra}	LOO RMSE (mg g^{-1})
0	364
10	335
20	336
30	319
50	321
100	285

Adding 100 interpolated points reduces RMSE from 364 to 285 mg g^{-1} (-22%), confirming that denser coverage — even with synthetic midpoints — yields measurable improvement. Real experimental data in the gap regions would be even more effective because it directly reduces the variance due to the high-gradient boundary.

E Consolidated Upper-Bound Analysis and Recommendations

E.1 Theoretical Minimum Achievable RMSE

Combining the topological and Nyquist analyses, we can estimate upper and lower bounds on the achievable RMSE:

1. **Hard lower bound (nugget):** The semivariogram nugget $\gamma(0) \approx 0$ indicates no irreducible measurement noise. In principle, RMSE could approach zero with sufficiently dense data.
2. **Data-density upper bound (k -NN oracle):** With the *current* 47 train+val samples, the best any interpolation model can achieve is $\text{RMSE} \approx 328 \text{ mg g}^{-1}$ (1-NN LOO). The best GB model achieves 311 mg g^{-1} (10-trial) or 368 mg g^{-1} (LOO), in either case close to the oracle.
3. **Connectivity constraint:** The feature-space graph requires $\varepsilon = 7.3$ to connect, meaning there are genuine voids where any model must extrapolate rather than interpolate. The isolated sample 30 (NN distance = 7.24) alone contributes an irreducible extrapolation error whenever it appears in a test set.
4. **Nyquist bound for target RMSE 100 mg g^{-1} :** Requires approximately 63 uniformly distributed samples (Table 20). The current 56 samples are non-uniformly distributed and several are isolated, so ~ 100 well-targeted additional experiments would be needed in practice.

E.2 Revised Data Collection Recommendations

Based on the topological and Nyquist analyses, Table 23 provides a prioritised sampling strategy guided by the principles of active learning [16] and space-filling experimental design [13, 15].

E.3 Expected Performance After Targeted Collection

Using the k -NN interpolation estimate (Equation 2) and the data augmentation simulation, the expected RMSE after adding 75 well-targeted experiments is approximately:

$$\text{RMSE}_{75} \approx 328 \times \left(\frac{56}{56 + 75} \right)^{1/d_{\text{eff}}} \approx 328 \times 0.85 \approx 180\text{--}220 \text{ mg g}^{-1} \quad (4)$$

This is consistent with the power-law projection from Section 6 and with the simulation result (100 interpolated points $\rightarrow$ RMSE 285 mg g^{-1}).

Table 23: Revised data collection strategy based on topological and Nyquist analysis, ordered by expected RMSE reduction per sample collected.

Region	$n_{\min}$	Rationale
SSA $\in [600, 1400]$, conc $\in [1, 4]$ mg/g (transition zone)	30	Bridges the two main clusters; directly addresses the high-gradient boundary ($f_{\max} = 1,461$); reduces kNN oracle bound by $\sim 50 \text{ mg g}^{-1}$ per 30 samples.
Target $y \in [532, 993]$ mg/g (decile deficit = +154)	20	Most under-sampled decile relative to spacing; filling this gap will close the topological void between mid-range and high-performance clusters.
Target $y \in [76, 145]$ and $[217, 357]$ mg/g (deficit >100)	20	Low-end transition; important for model accuracy on realistic operational conditions.
Neighbours of Sample 30 (extremal outlier)	5	Sample 30 sits in an NN-void of radius 7.24; 5 experiments near its feature values would reduce its contribution to extrapolation error.
Total minimum	75	

Part I

Dataset 5 — Revised Feature Set

This part repeats the analysis of Part I on a revised version of the methylene blue (MB) dye adsorption dataset (`updated_dataset_5_madam_MB_dye.xlsx`). The revised dataset drops the “Equilibrium time” feature and renames columns, yielding $N = 48$ observations with 6 input features and the same adsorption capacity target (“QM mg/g Y”). All methodology, code infrastructure, and statistical tests are identical to Part I; only the data change.

F Dataset 5 Overview

F.1 Descriptive Statistics

Compared with Dataset 4 (Section 2), the sample size decreased from 56 to 48, “Equilibrium time” was removed, and the “Molar dye/AC concentration” feature was replaced by a “Dosage ratio” expressed in g L^{-1} units. The target distribution is structurally similar (mean 436 vs. 479, std 518 vs. 542, max 2251 in both).

F.2 Skewness and Outliers

The target skewness (2.27) is slightly higher than in Dataset 4 (2.13). Approximately 80 % of samples have $QM \leq 595 \text{ mg g}^{-1}$, and the top 10 % exceeds 936 mg g^{-1} (only 5 samples).

Table 24: Descriptive statistics — Dataset 5 ($N = 48$, 6 features).

Feature	Mean	Std	Min	Q1	Median	Q3	Max
SSA ($\text{m}^2 \text{g}^{-1}$)	1031	763	1.6	492	896	1547	2869
PV ($\text{cm}^3 \text{g}^{-1}$)	0.64	0.46	0.001	0.35	0.60	0.84	2.05
MPD (nm)	3.53	3.76	0.74	1.89	2.43	3.31	23.4
Dosage ratio ($\text{g L}_{\text{dye}}^{-1} / \text{g L}_{\text{AC}}^{-1}$)	0.59	0.86	0.003	0.10	0.26	0.57	4.29
pH	8.50	1.87	6.0	7.0	8.0	11.0	12.0
Temperature ($^{\circ}\text{C}$)	29.3	7.43	20	25	27.5	30	55
QM (mg g^{-1})	436	518	13	133	277	458	2251

Table 25: Skewness and IQR outlier counts — Dataset 5.

Feature	Skewness	Outliers (IQR)	Interpretation
SSA	0.82	0	Approximately symmetric
PV	1.16	2	Right-skewed
MPD	3.86	5	Extremely skewed
Dosage ratio	2.54	6	Right-skewed
pH	0.33	0	Nearly symmetric
Temperature	1.99	5	Clustered at 25–30 $^{\circ}\text{C}$
Target (QM)	2.27	5	Strongly right-skewed

G Dataset 5 — Model Selection and HPO

G.1 Quantitative Dataset Properties

Table 26: Quantitative dataset properties — Dataset 5 ($N = 48$, $d = 6$).

Property	Value	Implication for modelling
Sample-to-feature ratio N/d	8.0	Same ratio as Dataset 4; remains very small
Target normality (Shapiro–Wilk p)	< 0.001	Non-Gaussian target (skew = 2.20); linear-residual models penalised
Features passing normality test	0/6	All features non-normal
High feature correlations ($ r > 0.7$)	1 pair	SSA–PV (0.77); similar to Dataset 4
Significant non-linear feature–target relations	2 of 6	PV, Dosage ratio have Pearson–Spearman gaps > 0.15
Total IQR outliers across features	18	$\sim 3.0/\text{feature}$ (reduced from 4.6 in Dataset 4)
Target range ratio (max/min)	173	Unchanged from Dataset 4

G.2 Model-Family Assumption Alignment

The alignment analysis follows the same framework as Section 3. With $N/d = 8$, non-normal features, and non-linear effects, the ranking remains: **Gradient Boosting** $\approx$

Random Forest > SVR $\gg$ Ridge > Lasso. The reduced outlier count (18 vs. 32) modestly favours all models, but the fundamental characteristics favour tree-based ensembles.

G.3 HPO Results

Table 27: HPO results — Dataset 5 (5-fold CV on 40-sample train+val pool).

Rank	Model	CV RMSE	CV R^2	10-trial RMSE	Best hyperparameters
1	Gradient Boosting	263.6	0.354	187.8 ± 75.7	$n_{\text{est}} = 30$, $\eta = 0.1$, depth=5, sub=1.0, mf=sqrt
2	Random Forest	283.8	0.326	211.5 ± 92.5	$n_{\text{est}} = 200$, depth=all, mf=all, leaf=1
3	SVR (RBF)	352.2	0.404	307.0 ± 115.3	$C = 500$, $\varepsilon = 0.01$, $\gamma=\text{scale}$
4	Ridge	372.1	-0.087	297.8 ± 108.5	$\alpha = 10$
5	Lasso	430.4	-0.668	355.8 ± 154.7	$\alpha = 10$

Interpretation.

1. **Gradient Boosting is again the best model**, with a 20 mg g^{-1} advantage over Random Forest on CV RMSE and 24 mg g^{-1} on 10-trial RMSE.
2. The **gap between tree ensembles and linear/kernel models is wider** than in Dataset 4: GB achieves $\text{CV } R^2 = 0.354$ vs. Ridge’s -0.087 . The dosage ratio feature appears to interact non-linearly with SSA more strongly in this dataset.
3. The HPO-selected configuration ($n_{\text{est}} = 30$, $\eta = 0.1$, max_depth=5, subsample=1.0, max_features=sqrt) uses a higher learning rate and deeper trees than Dataset 4, reflecting that the revised feature set supports a more expressive model.

G.4 Selected Model Configuration

```
GradientBoostingRegressor(
    n_estimators = 30,
    learning_rate = 0.1,
    max_depth = 5,
    subsample = 1.0,
    max_features = 'sqrt',
    min_samples_split = 2,
    min_samples_leaf = 1,
    random_state = <trial seed>
)
```

H Dataset 5 — Evaluation: 10-Trial Random Splits

H.1 Protocol

The same protocol as Section 5 is used: the 48 samples are split into train+val ($n = 40$) and fixed held-out test ($n = 8$) by `prepare.py`, then the 40-sample pool is re-split 10

times (80/20) with the same seeds.

H.2 Per-Trial Performance

Table 28: Per-trial metrics for the HPO-selected Gradient Boosting model on Dataset 5. “Frac. high” = fraction of test samples with target > 90 th percentile ($1\,051\text{ mg g}^{-1}$).

Seed	n_{train}	n_{test}	Test R^2	Test RMSE	Frac. high
521	32	8	0.948	52.7	0.00
737	32	8	-0.362	144.3	0.00
740	32	8	0.760	311.6	0.25
660	32	8	0.888	182.7	0.12
411	32	8	0.925	130.6	0.12
678	32	8	0.705	170.4	0.00
626	32	8	0.792	293.4	0.12
513	32	8	0.849	207.0	0.12
859	32	8	0.543	214.0	0.00
136	32	8	0.882	171.4	0.12
Mean			0.693	187.8	
Std			0.411	75.7	
CV			0.593	0.403	

H.3 Quantitative Diagnosis

Observation 1: Improved mean but comparable variance. Mean RMSE dropped from 311.0 (Dataset 4) to 187.8 mg g^{-1} (Dataset 5), a 40 % reduction despite the smaller sample size ($n_{\text{train}} = 32$ vs. 37). However, $\text{CV}_{\text{RMSE}} = 0.403$ remains similar to Dataset 4’s 0.411, and $\text{CV}_{R^2} = 0.593$ is somewhat higher (driven by the negative- R^2 trial at seed 737). The 95 % bootstrap CI for mean RMSE is $(143.0, 232.4)\text{ mg g}^{-1}$, spanning $\pm 24\%$ of the mean.

Observation 2: Test composition effect is weaker. Trials where no high-value samples appear in the test set (seeds 521, 737, 678, 859) show mixed performance — some very good (521: $\text{RMSE}=52.7$) and some poor (737: $\text{RMSE}=144.3$ with $R^2 = -0.362$). This contrasts with Dataset 4 where the presence of high-value test samples was the dominant driver of RMSE variance. The revised feature set produces more uniform prediction quality across the target range.

Observation 3: Near-independent test folds. The test-set Jaccard similarity is 0.115 ± 0.091 , close to Dataset 4’s 0.123 ± 0.073 and to the theoretical minimum, confirming independent evaluation folds.

Observation 4: Training overfitting persists. Training $R^2 \in [0.999, 1.000]$ across all trials (near-perfect fit), while test R^2 fluctuates from -0.362 to 0.948 . The overfitting gap is driven by data scarcity, not model complexity.

I Dataset 5 — Statistical Comparison: 10-Trial vs. LOO-CV

Table 29: Evaluation methodology comparison — Dataset 5. Both protocols report in original mg g^{-1} space.

Criterion	10-Trial 80/20	Leave-One-Out CV
Training fold size	≈ 32 of 40 (80 %)	39 of 40 (97.5 %)
Test fold size	8 samples	1 sample
Metric space	Original mg g^{-1}	Original mg g^{-1}
RMSE	187.8 ± 75.7	220.8
R^2	0.693 ± 0.411	0.838

Summary. LOO-CV yields $\text{RMSE} = 220.8 \text{ mg g}^{-1}$ and $R^2 = 0.838$ in original space — a higher RMSE than the 10-trial mean (187.8) but with a notably better R^2 (0.838 vs. 0.693). The LOO-CV R^2 being higher is consistent with the larger training fold (39 vs. 32 samples); the higher RMSE reflects the sensitivity of single-sample test folds to isolated high-gradient regions.

Compared with Dataset 4 (LOO RMSE 368.0, R^2 0.514), the revised feature set shows a substantial improvement in both metrics despite having fewer total samples.

J Dataset 5 — Learning Curve Analysis

J.1 Empirical Learning Curve

Table 30: Learning curve — Dataset 5 (100 random trials per size, testing on all remaining samples from the 40-sample pool).

n_{train}	Mean RMSE (mg g^{-1})	Std RMSE	CV
12	374	138	0.37
15	345	131	0.38
18	328	127	0.39
22	316	131	0.41
26	277	121	0.44
28	266	106	0.40
32	226	97	0.43

J.2 Power-Law Fit

$$\text{RMSE}(n) \approx 1233.9 \cdot n^{-0.465}, \quad r = -0.957 \quad (5)$$

The exponent $b \approx -0.47$ is markedly steeper than Dataset 4’s $b \approx -0.17$, approaching the root- n regime ($b = -0.5$). This indicates that Dataset 5 is **not yet saturated**: each additional sample still yields a meaningful reduction in RMSE.

Table 31: Projected training set size to achieve target RMSE — Dataset 5 (power-law extrapolation).

Target RMSE (mg g ⁻¹)	n_{train} required
250	≈ 31
200	≈ 50
150	≈ 93
100	≈ 222

Compared with Dataset 4 (which required ≈ 98 samples for $\text{RMSE} \leq 300$), Dataset 5 reaches the same threshold at only ≈ 22 samples — reflecting the steeper learning exponent.

K Dataset 5 — Feature Importance and Domain Coverage

K.1 Feature Importances

Table 32: Gradient Boosting feature importances — Dataset 5 (full 40-sample pool, HPO-selected model from Section G.3).

Feature	Importance
SSA (m ² g ⁻¹)	0.3911
Dosage ratio	0.3444
PV (cm ³ g ⁻¹)	0.1497
MPD (nm)	0.1066
Temperature (°C)	0.0400
pH	0.0098

SSA is now the dominant feature (importance 0.39 vs. 0.30 in Dataset 4), and the top two features (SSA, Dosage ratio) account for 74 % of total importance. pH has negligible importance (0.010), consistent with the near-zero Spearman correlation ($\rho = -0.07$, $p = 0.636$).

K.2 Spearman Correlations

SSA’s Spearman correlation increased from 0.766 (Dataset 4) to 0.820, while PV increased from 0.594 to 0.651. Temperature and pH remain statistically non-significant.

K.3 Quadrant Analysis (SSA vs. Dosage Ratio)

The imbalance is qualitatively similar to Dataset 4: off-diagonal quadrants (15 samples, 31 %) are under-represented relative to the diagonal (33 samples, 69 %). The high-SSA/high-Dosage quadrant (std 695) again exhibits the widest dispersion.

Table 33: Spearman rank correlations with the target — Dataset 5.

Feature	ρ	p -value
SSA	0.820	< 0.001
PV	0.651	< 0.001
Dosage ratio	0.571	< 0.001
MPD	-0.336	0.020
Temp	0.194	0.186
pH	-0.070	0.636

Table 34: Sample density and mean target in SSA–Dosage quadrants (split at medians: $\text{SSA}_{\text{med}} = 896$, $\text{Dosage}_{\text{med}} = 0.26$).

Region	n	Mean target (mg g^{-1})
Low SSA / Low Dosage	17	157 ± 100
High SSA / Low Dosage	7	523 ± 353
Low SSA / High Dosage	8	169 ± 111
High SSA / High Dosage	16	827 ± 695

K.4 Target Distribution by Decile

Table 35: Samples per target decile — Dataset 5.

Range (mg g^{-1})	n	Note
13 – 42	5	
42 – 90	5	
90 – 194	5	
194 – 223	4	
223 – 277	5	
277 – 303	5	
303 – 398	4	
398 – 595	5	
595 – 936	5	
936 – 2251	5	Tail: 5 samples over 100$\times$ range

The decile structure is broadly similar to Dataset 4, with two thin deciles (194–223 and 303–398) having only 4 samples each. The top decile spans 936–2251 mg g^{-1} — a factor of 2.4.

L Dataset 5 — Topological Analysis

L.1 Intrinsic Dimensionality

The intrinsic dimensionality ($d_{\text{eff}} \approx 3.0$) is slightly higher than Dataset 4’s 2.9 despite having one fewer ambient feature (6 vs. 7). This suggests that the removed “Equilibrium time” feature was largely redundant.

Table 36: Intrinsic dimensionality estimates — Dataset 5.

Method	Estimate	Notes
PCA (90 % variance threshold)	5 PCs	87.8 % at 4 PCs
Two-Nearest-Neighbour (2NN)	3.07	
Correlation dimension	2.96	$r = 0.994$

L.2 Connectivity and Voids

Table 37: Vietoris–Rips connectivity — Dataset 5 (standardised feature space).

ε	Components
0.5	45
1.0	20
1.5	11
2.0	9
2.5	5
3.0	3
4.0	1

Full connectivity is reached at $\varepsilon \approx 4.0$ — substantially lower than Dataset 4’s 7.3. This confirms that the revised dataset has fewer extreme voids in the feature space.

L.3 Isolated Samples

Samples with nearest-neighbour distance $> 2 \times \text{median}$ ($2 \times 0.882 = 1.76$) are listed in Table 38.

Table 38: Isolated samples — Dataset 5 (NN distance > 1.76).

Index	y (mg g ^{−1})	NN distance	Significance
43	220	3.42	Most isolated ; nearest sample 3.42 units away
6	1684	3.12	High-performance isolated point
5	16	2.91	Low-performance isolated point
11	980	2.86	High-mid isolated point
1	195	2.46	Mid-range isolated point
30	644	2.23	Mid-range isolated point
7	2080	2.22	High-performance isolated point
15	269	1.91	Mid-range isolated point
40	916	1.79	High-mid isolated point

Nine samples (18.8 %) are isolated — higher than Dataset 4’s 7 (12.5 %). However, the maximum NN distance is 3.42 (vs. 7.24), indicating that while more samples sit in local voids, no single sample is as extremely isolated as Dataset 4’s sample 30.

M Dataset 5 — Nyquist Analysis

M.1 Spatial Frequency

Using all $\binom{48}{2} = 1,128$ sample pairs in standardised feature space:

- Median gradient: $102.7 \text{ mg g}^{-1}/\text{unit}$
- 95th-percentile gradient: $507 \text{ mg g}^{-1}/\text{unit}$
- Maximum gradient: $1,657 \text{ mg g}^{-1}/\text{unit}$

The median spatial frequency is lower than Dataset 4’s 139.4, reflecting a smoother target response surface on average.

M.2 Semivariogram

Table 39: Experimental semivariogram — Dataset 5. Sill = $\text{Var}(y) = 263,003 \text{ (mg g}^{-1})^2$, $\sigma_y = 513 \text{ mg g}^{-1}$.

Distance h range	$\gamma(h)$	$\gamma(h)/\text{sill}$
[0.23, 1.36)	19 519	0.07
[1.36, 1.80)	24 415	0.09
[1.80, 2.22)	67 336	0.26
[2.22, 2.56)	111 534	0.42
[2.56, 2.99)	202 879	0.77
[2.99, 3.41)	332 715	1.27
[3.41, 3.93)	450 057	1.71
[3.93, 4.38)	527 843	2.01
[4.38, 5.33)	395 461	1.50
[5.33, 8.73)	547 882	2.08
Nugget $\gamma(0)$	$\approx 21,283$	0.08
Sill	263 003	1.000
Range (80 % sill)	$h \approx 2.50$	—

The nugget is approximately 21 000 (8 % of the sill), compared with near-zero in Dataset 4. This small but non-negligible nugget suggests a modest level of measurement noise or micro-scale variability, implying that RMSE cannot approach zero even with arbitrarily dense sampling. The variogram range ($h \approx 2.50$) is similar to Dataset 4’s 2.37.

M.3 k -NN Oracle

The best oracle ($k = 3$) achieves $\text{RMSE} \approx 310 \text{ mg g}^{-1}$. The GB model’s 10-trial mean (187.8) is *better* than the k -NN oracle, which is expected: parametric models can extrapolate beyond the nearest-neighbour boundary, while k -NN is purely interpolative. The LOO-CV RMSE of the GB model (220.8) is also better than the oracle, confirming that the parametric model is extracting structural information (not just local averaging).

Table 40: k -NN Oracle LOO-CV RMSE on the 40-sample train+val pool — Dataset 5 (original mg g^{-1} space).

k	LOO RMSE (mg g^{-1})
1	328
2	339
3	310
5	338
7	380
10	407

Table 41: Nyquist sampling requirement — Dataset 5.

Target RMSE (mg g^{-1})	Required spacing δ	n_{required}
300	5.84	3
200	3.89	12
150	2.92	29
100	1.95	100
50	0.97	839

M.4 Nyquist Requirements

Using $d_{\text{eff}} = 3.07$, $\bar{f} = 102.7$, and $L = 8.73$:

The Nyquist estimate for $\text{RMSE} \leq 100$ requires ~ 100 uniformly distributed samples, consistent with the power-law projection of ≈ 222 training samples (the excess accounts for non-uniformity of the current distribution).

M.5 Data Augmentation Simulation

Table 42: k -NN ($k = 3$) LOO RMSE on original 48 samples after adding interpolated synthetic training points — Dataset 5.

n_{extra}	LOO RMSE (mg g^{-1})
0	288
10	288
20	284
30	247
50	172
100	109

Adding 100 interpolated points reduces the k -NN oracle RMSE from 288 to 109 mg g^{-1} (-62%), a steeper improvement than Dataset 4’s -22% with the same augmentation count.

N Dataset 5 — Recommendations

N.1 Per-Decile Sampling Deficit

Table 43: Intra-decile density analysis — Dataset 5. Deficit estimated via $n_{\text{extra}} = n_{\text{current}} ((\delta/\delta_{\text{med}})^{d_{\text{eff}}} - 1)$, where $\delta_{\text{med}} = 0.882$.

Target decile (mg g^{-1})	n	Spacing	Deficit
[13, 42)	5	1.67	+30
[42, 90)	5	1.41	+16
[90, 194)	5	1.12	+5
[194, 223)	4	3.31	+227
[223, 277)	5	1.54	+22
[277, 303)	5	1.71	+33
[303, 398)	4	1.44	+14
[398, 595)	5	0.99	+2
[595, 936)	5	1.71	+32
[936, 2251]	5	1.85	+43
Total deficit			+424

The most under-sampled decile ([194, 223), deficit +227) is a narrow range with only 4 samples spread over 3.31 spacing units — the highest ratio in either dataset. This appears to be a transition region between low-performance and mid-performance adsorbents.

N.2 Revised Data Collection Strategy

Table 44: Recommended data collection strategy — Dataset 5.

Region	n_{min}	Rationale
SSA $\in [600, 1400]$, dosage $\in [0.3, 1.5]$ (transition zone)	20	Bridges the diagonal clusters; fills the high-gradient boundary.
Target $y \in [194, 223]$ and $[303, 398]$ (thin deciles)	15	These 4-sample deciles have the worst spacing and drive interpolation error.
Target $y \in [595, 936]$ and $[936, 2251]$ (high-performance tail)	15	Total deficit = +75; high-value region most relevant to applications.
Neighbours of sample 43 and sample 5 (isolated extremes)	5	Reduce NN-void contribution to extrapolation error.
Total minimum	55	

Table 45: Phased data collection roadmap — Dataset 5.

Phase	Action	Additional n	Projected RMSE
Current	Baseline dataset	48 total	188 ± 76
Phase 1	+20 transition zone, +10 thin deciles	+30	$\approx 150\text{--}170$
Phase 2	+15 high-tail, +5 isolated neighbours	+20	$\approx 120\text{--}140$
Phase 3	Stratified augmentation	+30	$\approx 90\text{--}110$
Phase 3 total		≈ 128	≈ 100

N.3 Phased Roadmap

O Dataset 5 — Summary and Comparison with Dataset 4

Table 46: Head-to-head comparison of Dataset 4 and Dataset 5.

Metric	Dataset 4	Dataset 5
N (samples)	56	48
d (features)	7	6
d_{eff} (intrinsic dim.)	2.9	3.07
Target mean $\pm$ std	479 ± 542	436 ± 518
Target skewness	2.13	2.27
10-trial mean RMSE	311.0	187.8
10-trial RMSE std	127.7	75.7
LOO-CV RMSE (original space)	368.0	220.8
LOO-CV R^2	0.514	0.838
k -NN oracle RMSE	328	310
Power-law exponent b	−0.166	−0.465
Variogram range (80 % sill)	2.37	2.50
Variogram nugget	≈ 0	$\approx 21,000$
ε for full connectivity	7.30	4.00
Max NN distance	7.24	3.42
SSA importance	0.299	0.391
Feature 1 Spearman ρ (SSA)	0.766	0.820

Key findings.

1. **Dataset 5 yields markedly better prediction performance** (−40 % RMSE, +63 % R^2 on LOO-CV) despite having 8 fewer samples and 1 fewer feature. This suggests that the “Equilibrium time” variable in Dataset 4 introduced noise or redundancy rather than useful signal.
2. **The learning curve is no longer saturated.** The steeper power-law exponent (−0.47 vs. −0.17) means additional data will produce substantial RMSE reductions.

Roughly 50 training samples are needed for $\text{RMSE} \leq 200$, compared with $\approx 1,124$ in Dataset 4.

3. **Feature-space topology is more compact.** Full connectivity at $\varepsilon = 4.0$ (vs. 7.3) and a maximum NN distance of 3.42 (vs. 7.24) indicate fewer extreme voids, enabling more reliable interpolation.
4. **SSA is the single most important feature** (importance 0.38, Spearman $\rho = 0.820$), followed by Dosage ratio (0.30). Together they account for 68 % of model importance.
5. **A non-zero variogram nugget** ($\approx 21,000$) appeared in Dataset 5, suggesting modest measurement noise that places a fundamental lower bound on achievable RMSE of approximately $\sqrt{21,000} \approx 146 \text{ mg g}^{-1}$ — or lower if the nugget is partly attributable to short-range spatial variability.
6. **Recommended minimum additional data: 55 targeted experiments** (Phase 1+2), with a goal of reaching $\text{RMSE} \approx 100 \text{ mg g}^{-1}$ at 128 total samples (Phase 3).

Appendix D: Dataset 5 Statistical Tests Summary

Table 47: Key statistical results — Dataset 5.

Metric	Value	Interpretation
Test R^2 mean $\pm$ std	0.693 ± 0.411	Very high variance
Test RMSE mean $\pm$ std	188 ± 76	$\text{CV} = 0.40$
LOO-CV RMSE (original space)	220.8	Near-unbiased, single estimate
LOO-CV R^2 (original space)	0.838	Substantially better than 10-trial mean
95 % Bootstrap CI (RMSE)	(143, 232)	± 24 % of mean
Test set Jaccard similarity	0.115 ± 0.091	Near-independent folds
Pearson(RMSE, test mean)	$r = 0.375$	Not significant ($p = 0.285$)
Pearson(R^2 , test max)	$r = 0.622$	Borderline ($p = 0.055$)
Power-law exponent	-0.465	Near root- n learning
Power-law fit r	-0.957	Good fit
Spearman(SSA, target)	$\rho = 0.820$	Strongest feature
Spearman(Dosage, target)	$\rho = 0.571$	Second strongest
Spearman(PV, target)	$\rho = 0.651$	Third strongest

The model configuration for Dataset 5 was selected via HPO (Section G.3): Gradient-BoostingRegressor with $n_{\text{est}} = 30$, $\eta = 0.1$, `max_depth=5`, `subsample=1.0`, `max_features=sqrt`.

Appendix C: Topological & Nyquist Summary Statistics

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Table 48: Key topology and Nyquist analysis results.

Quantity	Value	Interpretation
Ambient dimensionality d	7	Measured features
Intrinsic dimensionality d_{eff}	2.9	2NN estimate
Correlation dimension	2.18	Power-law fit on pairwise dist
PCA 90 % variance	5 PCs	Remaining 2 PCs carry <10 %
Median NN distance	0.806	Normalised feature space
Maximum NN distance	7.24	Sample 30 is an extremal outlier
ϵ for full connectivity	7.30	vs. median 0.81; factor 9
Isolated samples ($> 2 \times \text{median}$)	7	12.5 % of dataset
Variogram range (80 % sill)	2.37	Correlation length of target
Variogram nugget	≈ 0	No irreducible noise
Median spatial frequency $\bar{f}$	139 mg g ⁻¹ /unit	
Max spatial frequency f_{max}	1 708 mg g ⁻¹ /unit	
k -NN oracle RMSE (best k)	328 mg g ⁻¹	Interpolation ceiling (1-NN LOO)
Best GB model RMSE (10-trial)	311 mg g ⁻¹	HPO-selected model
Best GB model RMSE (LOO-CV)	368 mg g ⁻¹	Near-oracle, same units
Augmented oracle (+100 pts)	285 mg g ⁻¹	Validates Nyquist prediction
Nyquist n for RMSE 100	~ 63 uniform	Idealised estimate
Practical n for RMSE 100	~ 130 targeted	Accounting for non-uniformity
Total sample deficit (decile)	636	For uniform median spacing

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